

Project Proposal

Project Objective

Modeling Ann Arbor temperature trends with least squares regression.

Application: Analyzing Ann Arbor Temperature Trends

We will apply the least squares method to discover the temperature of Ann Arbor. The goal of this application is to determine if there has been a change in Ann Arbor's climate over the past few decades. We will use past weather data to visualize weather trends. By using the least squares model, we will provide a mathematical model for the data. With this model, we aim to describe historical trends and explore possible future temperature patterns.

Dependent Variable (Y)	Average daily temperature in °F, calculated from daily high and low temperatures.
Independent Variables (X)	Month of the Year: primary predictor for seasonality. Sine and cosine terms for cyclical seasonal variations.

We plan to fit multiple models to the data and compare their performance, performing separate analyses for each decade to observe long-term trends in temperatures. We are expecting a rising trend in temperature over the decades due to global warming.

Harmonic Regression

The *harmonic regression model* with a linear trend is given by

$$T = \beta_0 + \beta_1 t + \beta_2 \sin\left(\frac{2\pi m}{12}\right) + \beta_3 \cos\left(\frac{2\pi m}{12}\right) + \epsilon,$$

where t represents time, capturing long-term temperature trends. m represents the month of the year. The sine and cosine terms model periodic seasonal variations. ϵ represents random error.

To allow for possible nonlinear long-term trends, we also consider a quadratic trend model with seasonal components

$$T = \beta_0 + \beta_1 t + \beta_2 t^2 + \beta_3 \sin\left(\frac{2\pi m}{12}\right) + \beta_4 \cos\left(\frac{2\pi m}{12}\right) + \epsilon.$$

Model Evaluation

Residual Minimization	We will apply the least squares method to minimize the difference between observed and predicted temperatures, aiming to develop a model that provides accurate temperature predictions.
Regression Analysis	Model performance will be assessed using R-squared and residual analysis.

Mathematical Theory

Projection Onto Subspaces: We view our independent variables as vectors that span a subspace (the Column Space of X). Since our observed temperature vector y does not likely lie within this subspace, we find the projection $\hat{y}$ that minimizes the distance to y , where $\hat{y} = \text{proj}_{\text{Col}(X)}(y)$.

Normal Equations

To find the coefficients $\hat{\beta}$ that define the projection by solving

$$X^T X \hat{\beta} = X^T y,$$

where $\hat{\beta} = (X^T X)^{-1} X^T y$.

QR-decomposition

QR-decomposition factors the design matrix X into $X = QR$, where Q has orthonormal columns and R is upper triangular. The least squares problem can then be solved by

$$R\hat{\beta} = Q^T y.$$

Implementation Plan

Data collection	We will be pulling data from <u>NOAA</u> (National Oceanic and Atmospheric Administration). The data has been uploaded on GitHub (https://github.com/QihangStudio/Ann-Arbor-Weather-Trend/blob/main/data.csv) in CSV file.
Programming	We will employ programming to visualize the trend in variables in order to select a mathematical model that will best fit the data. Due to the complexity and high-dimensional data, we will not be able to do least squares regression by hand. Therefore, we will also utilize programming tools to do the math.
Visualization	We will be plotting the data points against the fitted curve and visualizing the residuals through programming.

Tools and Libraries

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn